Emanuel André Medina Araujo

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Education

Technical University of Munich — Oct. 2026 – Present
Master of Science in Informatics — *Munich, DE*

Technical University of Munich — Oct. 2020 – Aug. 2026
Bachelor of Science in Computer Science, Application Area in Physics — *Munich, DE*

Technical University of Munich — Oct. 2024 – Sep. 2025
Parallel studies in Physics (B.Sc. programme, not completed) — *Munich, DE*

Herzog-Christoph-Gymnasium — Sep. 2012 – Sep. 2020
Abitur, Awarded MINT-EC certificate with distinction — *Beilstein, DE*

Experience

Working Student Software Engineer — Nov. 2024 – Present
Vector Informatik GmbH - VSceneCreator (3D OPENDrive Editor) — *Munich, DE*

- Helped ship the 0.1 to 0.4 alpha versions of VSceneCreator (Unity/C#) through 43 merged pull requests, suggesting, designing and implementing features in close collaboration with Product Management.
- Built a polyline trajectory editing mode with OpenSCENARIO load and save (Unity/C#): data model, controllers, inspectors, validation issues, and a lossless round-trip of optional XML attributes.
- Designed geometry simplification for OpenDRIVE road networks: detects consecutive line and arc segments with matching heading and curvature and merges them through single or bulk fixes, with undo/redo, live validation and PlayMode tests.
- Cut CI time for editor-only pull requests from 29.8 to 18.8 min and from 43.9 to 33.7 min on two build nodes (Jenkins) by skipping unrelated packaging, reusing module results within a run, and retrying flaky tests in place.
- Built a read-only Explore Mode, the default without a licence: hover readout of world and road (s/t) coordinates, an overview tree of the whole network, and a 1,200-line PlayMode test suite; later extended the readout to every edit mode through a central collider manager.
- Reduced edit-mode switching delays on large networks (about 2 min for a 40 MB file) by creating editing handles on selection instead of up front; co-developed a staged loading pipeline with global progress.
- Extended OpenDRIVE import to convert single-sided lane sections into valid two-sided ones and fixed lane links when splitting sections, verified with file round-trip tests.
- Migrated the editor's geometry and road-network code from custom vector types to Unity.Mathematics double-precision types, with extension utilities and adapted tests.
- Ported three native C/C++ libraries to Linux with a platform-neutral export macro and a Conan setup that installs Windows and Linux plugins, so the Linux player gets its native dependencies.
- Set up the team's Rider integration, AI-assistant instruction files and MCP configuration docs; added auto-save with size and retention limits and a keyboard-shortcut system.

Working Student Software Engineer — Mar. 2021 – Oct. 2024
Vector Informatik GmbH - DYNA4 (Simulation environment for virtual driving tests) — *Munich, DE*

- Merged 218 pull requests across nine releases (6.0 to 9.1) in Java and Eclipse RCP, working directly with Product Management.
- Engineered a new editor and file type for signal aliases and favourites: filterable tree, save/load, database sync, a service layer and a migration handler for existing projects.
- Built reference analysis so users save only the files a database sync actually references; co-developed warning markers and a save quick fix for simulations with unsaved parameter files, and removed a deadlock risk by no longer querying the UI from background jobs.
- Extended the standalone run-package export from scenarios to tasks, refactoring the simulation runner to export instead of simulate.
- Co-developed automatic project migration: incompatible projects are detected and closed at startup and migrated in a background job by a new migration service, with a project backup step and JUnit plugin tests.
- Fixed internal errors when renaming the active project by replacing custom rename and delete actions with Eclipse refactoring participants.
- Added a guided clean-up of unreferenced project files on a new file-reference service, and a multi-axis signal plot that groups signals by physical unit (JavaFX).

- Reworked the trace signal view, including full signal paths, search over comments and clearer guidance when no signals are available.
- Authored 50+ test pull requests (SWTBot, JUnit), including reusable wrappers for an in-house form framework and a JUnit formatter that feeds CI results into test management (Xray).
- Improved maintainability by clearing IDE problems across 861 files: deprecated APIs, generics and externalised strings.

Projects

LoveNeverDies | *Legal Loves Tech Hackathon – Bavarian Ministry of Justice Challenge* Aug. 2026

- Built an end-to-end encrypted estate assistant for bereaved families (React, TypeScript, Supabase) in ten days with a team of lawyers, guiding users through inheritance deadlines via a 141-node legal decision tree.
- Designed a zero-trust architecture in which the server only stores ciphertext: hybrid post-quantum key exchange and signatures (ML-KEM + X25519, ML-DSA + Ed25519), AES-256-GCM, and a Shamir-secret-shared vault.
- Verified row-level security with 130 Vitest files against the real Postgres migrations (PGlite/WASM) and multi-user Playwright end-to-end tests; enforced the layered architecture through ESLint rules.

SchafPlay | *Peer-to-peer multiplayer card game (PWA)* Jun. 2026 – Aug. 2026

- Built an offline-first PWA for the Bavarian card game Schafkopf (React, TypeScript) with serverless peer-to-peer multiplayer over WebRTC data channels, using a manual SDP exchange instead of a signaling server.
- Implemented a host-authoritative game engine with per-player state redaction, so opponents' hands never go over the wire, plus heuristic AI opponents and a trick-by-trick replay analysis.
- Covered pairing, reconnect, and rule enforcement with a seeded Playwright end-to-end suite in GitHub Actions CI.

Autonomous AI Agent Orchestration (Hermes) | *Headless Server Infrastructure & Security* May 2026 – Present

- Architected and deployed a headless orchestration server on repurposed hardware running Ubuntu Server Pro, enforcing strict privacy via LUKS full-disk encryption, SSH key-based authentication, and Tailscale network isolation.
- Configured and launched the Hermes autonomous AI agent (Nous Research), leveraging its native Bitwarden Secrets Manager integration for secure, centralized credential handling.
- Engineered a real-time secure messaging gateway utilizing `signal-cli` wrapped in systemd services and D-Bus for agent notifications.

6D Object Pose Estimation for Robotics | *Robotics Seminar Paper* Apr. 2026 – Jul. 2026

- Wrote an IEEE-format survey of 6D object pose estimation, from classical geometric registration to zero-shot foundation-model methods, evaluated against generalization, real-time capability, and accuracy.
- Showed that no current method meets all three requirements and identified hybrid architectures (slow zero-shot initialization plus fast local tracking) as the most viable direction.

ClarissaApp | *Privacy-first personal health mobile app* Aug. 2025 – Feb. 2026

- Built a privacy-first mobile app for menstrual cycle tracking with React Native (Expo) and TypeScript.
- Implemented calendar-based cycle logging with symptom tracking and three themes (dark/light/red).
- Engineered offline-first storage with expo-sqlite, modular components/hooks, and a clear UX for fast daily logging.

Full-Stack ISP Comparison | *Check24 Scholarship Challenge* May 2025 – Jun. 2025

- Implemented a modular full-stack application (React + Vite + Tailwind frontend, Node.js/Express backend).
- Integrated timeout/health handling and an ISP provider normalization layer to produce consistent comparisons.
- Developed a share results feature with via a TinyURL export persisting the results within the link.

Learning Based Inverse Kinematics | *Bachelor Thesis* Jun. 2024 – Nov. 2024

- Designed and implemented learning-based IK solvers for 2- and 3-DOF manipulators in Python and Pytorch, comparing direct-angle regression with probabilistic outputs across MLP, LSTM, hybrid MLP+LSTM, and PPO.
- Built a synthetic data generation and evaluation pipeline; ran controlled experiments on a 10k-sample test set to quantify accuracy, convergence, and runtime across architectures and output parameterizations.
- Found that a simple MLP with direct-angle prediction consistently outperformed more complex models, highlighting a non-trivial complexity-performance trade-off and yielding clear guidance for practical IK model selection.

Ironman | *3.8km swim, 180km bike, 42.2km run; Overall Time : 14:21:26h* Oct. 2022 – Aug. 2024

- Completed full Ironman triathlon through disciplined training and a goal-oriented race plan, demonstrating resilience and on-the-fly problem-solving under fatigue and variable race conditions to reach the finish.

Internet Lab 2 | *Advanced Practical Networking Course* Oct. 2023 – Mar. 2024

- Completed course labs on advanced networking topics in a two-person team, building strong proficiency in GNU/Linux CLI and Cisco IOS for complex network configurations.
- Co-created a full WebAuthN lab from scratch (lecture + prelab + hands-on) covering passkeys, FIDO2/WebAuthN flows, and security keys, enabling peers to configure WebAuthN end-to-end.
- Engineered the practical stack with Keycloak and a Spring Boot demo app, guiding students through server setup, WebAuthN policy configuration, and packet analysis to deepen protocol understanding.

Cyber-Physical Systems Survey | *Advanced Cyber-Physical Seminar* Apr. 2023 – Jul. 2023
- Produced an in-depth research survey of numerical inverse kinematics in cluttered environments for robots.
- Evaluated Jacobian-based, heuristic, and meta-heuristic methods, for convergence, robustness, and performance.

FreshFinder | hackaTUM - HelloFresh Challenge Nov. 2023
- Developed a React + TypeScript web app for collaborative, parallel cooking sessions with a team of 4
- Designed a recipe dependency-graph architecture for task distribution using Supabase Realtime.
- Integrated the OpenAI API to dynamically adapt recipes for individual preferences and amounts.

StreamFindA | hackaTUM - Burda Challenge Nov. 2022
- Built a React front end with a Django/Python backend for swipe-based movie/series recommendations.
- Implemented a recommendation algorithm based on pairwise choices using dwell time and negative feedback.
- Parsed and normalized the provided multi-platform catalog to a usable format to power reliable recommendations.

Computer Architecture Practical Course | *Advanced Practical Course* Apr. 2021 – Jul. 2021
- Implemented iterative space-filling Z-curve generators in C and Assembly with SVG export and performance benchmarks.

Humanoid-controlled robot arm | *Student Engineering Academy* Sep. 2018 – Jul. 2019
- Programmed a self-built robot arm controlled via a humanoid interface.
- Developed a virtual movement visualization of the arm sleeve with flex sensors in Blender with Python.
- Programmed Arduino microcontroller in C/C++ to read movement data from the arm sleeve, transmit them wirelessly via radio using the SPI protocol to the robot arm and map them to the servo motors.

MATLAB physics visualizations | *International Summer Physics Institute, University of Notre Dame* Jul. 2018
- Completed Cosmic Ray Detection team project by building a detector plus analyzing and visualizing the data.
- Visualized and analyzed CERN CMS particle-physics data using MATLAB.

Technical Skills

Languages: C#, Java, TypeScript, JavaScript, Python, C/C++, SQL, MATLAB
Frameworks: React, React Native, Unity, PyTorch, Eclipse RCP, EMF, JavaFX, Supabase/PostgreSQL, WebRTC
Developer Skills: Git, Linux, GitHub Actions, Jenkins, Conan, SWTBot, JUnit, Playwright, Vitest, Jira, OpenDRIVE, OpenSCENARIO

Further Highschool Experience

STEM Programs and Workshops 2017 – 2019
- MINT-EC Workshop: Masterclass Particle Physics (Jun. 2019).
- Fraunhofer Talent Take Off - Vernetzen (2019).
- Regional MINT-Camp “Life on Mars” (May–Jun. 2019): Developed concepts for life-essential resources.
- Tag der Kombinatorik — group-based local math contest (Feb. 2019).
- Student Engineer Academy — project track (2018–2019).
- Fraunhofer Talent School (Mar. 2018): Implemented an image classifier (cats vs. dogs).
- Student Engineer Academy — Programming in C (2017–2018).

Early student Internships 2015 – 2019
- Bosch, Stuttgart-Feuerbach — voluntary engineering internship (Apr., 2019).
- Palfinger, Ilsfeld — voluntary mechanic internship (Sep. – Oct., 2017).
- Getrag, Untergruppenbach — voluntary mechanic internship (Sep., 2017).
- Vector, Stuttgart — “Mitmachen Ehrensache” social one-day placement (Dec., 2015).
- Eberhard Schweizer Grundschule Ganztagesgruppe — social week-long placement in nursery school (Nov., 2015).

Music and Community 2009 – 2020
- Leadership: Youth spokesman, Heinriet Sports Club (2016–2020); school mediator for grades 5–6 (2016–2020).
- Awards: D2 (2017, excellent–very good); Jugend musiziert 2nd prize (2015); D1 (2014, very good); scholarship (2013) and gifted promotion (2012) in trombone.
- Piano and trombone (advanced); District youth orchestra Heilbronn (2017–2020); Musikverein Heinriet (2014–2020); school band 1st trombone (2012–2014).
- Theater: Castle theater productions as trombonist and actor (2017: Das kleine Gespenst; 2016: Some like it hot).